

Skills Summary

Reading the Evidence for Triangle Congruence

Use this reference while completing your worksheet or when studying.

Skill 1 — Naming the criterion from the marks

- **SSS** — three pairs of sides.
- **SAS** — two sides and the angle *between* them.
- **ASA** — two angles and the side *between* them.
- **AAS** — two angles and a side *not* between them.
- **HL** — right triangles only: hypotenuse and one leg.

Not a criterion: SSA. Two sides and a non-included angle leave the second side free to swing to two closing points. HL is the one case rescued, because the right angle removes the second position.

Example: Two right triangles have hypotenuse $c = 13$ cm and one leg $a = 5$ cm, so they are congruent by HL. Find the remaining leg.

Step 1. HL names the two known parts as the hypotenuse and a leg, so the third side comes from the Pythagorean theorem rather than from any congruence rule.

Step 2. Remaining leg $= \sqrt{c^2 - a^2} = \sqrt{169 - 25} = \sqrt{144}$.

$\Rightarrow$ **12 cm**

Try it: Two right triangles have hypotenuse 25 cm and one leg 7 cm. Find the remaining leg.

check: 24

Skill 2 — From a criterion to an equation

Once two triangles are congruent, **every** pair of corresponding parts is congruent. So a labelled side matches the side in the same position of the other triangle, and a labelled angle matches the angle in the same position.

Read the correspondence off the congruence statement: in $\triangle ABC \cong \triangle DEF$, A matches D , B matches E , C matches F — so $\overline{AB}$ matches $\overline{DE}$ and $\angle B$ matches $\angle E$. Set the two expressions equal and solve.

Example: $\triangle ABC \cong \triangle DEF$ with $AB = 4x + 3$ cm and $DE = 2x + 15$ cm. Find x .

Step 1. The congruence statement pairs A with D and B with E , so $\overline{AB}$ and $\overline{DE}$ are corresponding sides and their lengths must be equal.

Step 2. $4x + 3 = 2x + 15$, so $2x = 12$.

$\Rightarrow$ **$x = 6$**

Try it: $\triangle ABC \cong \triangle DEF$ with $BC = 5x - 2$ cm and $EF = 3x + 14$ cm. Find x .

check: $8 = x$

Skill 3 — Finding the evidence nobody marked

Three pieces of evidence carry no tick marks and have to be read off the figure:

- a **shared side or angle** — congruent to itself by the *reflexive property*;
- **vertical angles** at a crossing — congruent by the Vertical Angles Theorem;
- a **third angle** — the angle sum of a triangle is 180° , so two matching angles force the third, which converts an AAS setup into ASA.

Example: Two angles of a triangle measure $a = 52^\circ$ and $b = 61^\circ$. Find the third angle.

Step 1. Two pairs of angles already match, so the third pair is free evidence — the angle sum fixes it without any extra given.

Step 2. Third angle $= 180 - a - b = 180 - 52 - 61$.

$\Rightarrow$ **67** degrees

Try it: Two angles of a triangle measure 48° and 74° . Find the third.

check: 58